

// What if a thought could move a limb?

CONTROL A PROSTHETIC HAND & LEG WITH YOUR MIND.

A non-invasive **EEG brain-computer interface** that reads motor-imagery brainwaves and drives a **3D-printed hand and leg in real time** — decoded by the **DB-ATCNet** neural network, accelerated on a **Zynq UltraScale+ FPGA**, and locked down with five-layer cryptographic security. Built for people who keep the intention to move, but have lost the means.

EEG • CZ
M/B RHYTHM

ETG 01



3D-printed robotic hand

20 KG·CM • 5 FINGERS • TENDON-DRIVEN

01 • SENSE
THINK
Imagine the movement — EEG captures it

02 • DECODE
DECODE
DB-ATCNet classifies the intent

03 • ACCEL
ACCELERATE
FPGA runs it in milliseconds

04 • ACT
MOVE
Servos grip & flex the limb

134 ms
Brain-to-motion latency
below the 200 ms reaction threshold

90 %
Live decoding accuracy
DB-ATCNet • HaLT • 5 electrodes

5 /19
Electrodes — down from 19
learned channel selection

♥ Millions living with **ALS, stroke, and spinal-cord injury** keep their intention to move — this interface gives them the means, with no muscle movement required.



/ 02

FIVE STAGES • ONE SIGNAL PATH

FROM THOUGHT TO MOVEMENT

01

SENSE

OpenBCI Cyton/Daisy records EEG — 16 channels at 250 Hz.

02

CONDITION

TLV9061 buffer + RC filter, then a **16-bit** ADS1115 ADC.

03

STREAM

ESP32-S3 windows the data in Q8.8 and streams it to the FPGA.

04

DECODE

DB-ATCNet classifies the motor-imagery intent on-chip.

05

ACTUATE

PCA9685 drives the servos — hand grips, ankle flexes.

AI

Why DB-ATCNet wins

Our model search ran from classic ML through modern deep nets — and a **Dual-Branch Attention Temporal Convolutional Network** came out on top, beating EEGNet, ATCNet & DeepConvNet.

Classic ML → EEGNet → CTNet → DB-ATCNet

Four stages: **temporal convolution** (μ/β rhythms) · **dual spatial branch** · **sliding windows** · **5-way ensemble voting**.

92.0%

→
91.5%

We swapped the heavy **Multi-Head Attention** for a hardware-efficient **CBAM** — a 0.5% accuracy trade for a model that actually fits an FPGA.

FIT

A lab cap → a headband

A clinical EEG cap needs 19 wet electrodes. A wearable needs **five dry ones you can self-apply in under two minutes**. Which five? Finding them meant searching **11,000+** combinations.

19 → 5

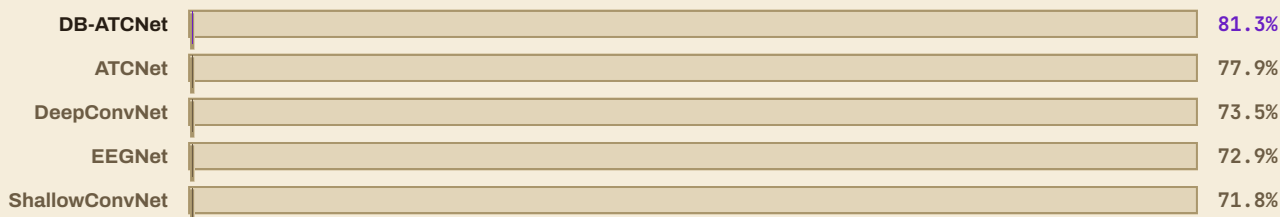
LEARNED SELECTION

Gumbel-softmax picked Cz · P3 · T5 · F7 · T3 — keeping **89.94%** accuracy on a fraction of the hardware.

The twist: weight-magnitude ranked Fp2 as important — but ablation proved **removing it improved performance**. What a network uses isn't always what it needs.

IT OUT-DECODES THE FIELD

PHYSIONET • 4-CLASS MOTOR IMAGERY • TEST ACCURACY



TRAINED & TESTED ON

PhysioNet EEGMMIDB • 109 subj

BCI Competition IV-2a • 4-class

BCI Competition IV-2b

HALT • 13 subj • LOSO

RONIIS
RESEARCH

Can we **fake fNIRS from EEG alone**? A diffusion model synthesised hemodynamic signals from raw EEG, lifting signal correlation **0.018 → 0.437** (a 24x jump) and recovering **71.7%** of the accuracy gain of real fNIRS hardware — at none of the cost or setup.



Validated across **131 subjects** with leave-one-subject-out cross-validation — then proven on physical, 3D-printed limbs.

/ 03

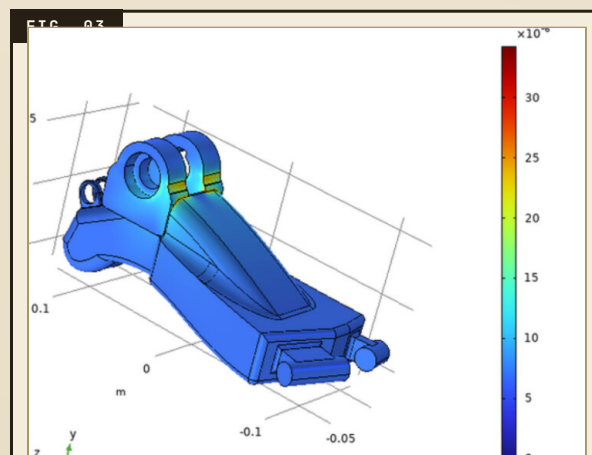
FPGA • COMSOL • LIVE DEMO

REAL HARDWARE, MEASURED.



The full system on the bench

ZCU106 • ESP32-S3 • Lipo • HAND + LEG



COMSOL stress study

PETG • FOS 1.6 @ 900 N

99.5%

FPGA DSP utilisation

1,719 / 1,728 DSP48E2 • ZCU106

2.31ms

Full-network inference

~4,300x faster than EEG arrival

0 fail

Timing endpoints failing

WNS +1.864 ns of 333,052

5.1w

Total on-chip power

~4.8 h FPGA battery runtime

90%

Live classification accuracy

HaLT • deployed 2-class system

89%

Inference completion rate

majority voter absorbs the rest

FIVE-LAYER SECURITY • SAFETY-FIRST CONTROL

+0.22% overhead

LAYER 1

AES-256-GCM
encryption

LAYER 2

HMAC-SHA-256 auth

LAYER 3

SHA-256 audit log

LAYER 4

Secure boot

LAYER 5

Bitstream encryption

Plus a **4-of-5 majority voter**, excitation persistence, a six-state safety FSM with a 1.5 s UART watchdog, and per-servo angle clamps & slew limits — so a glitch never reaches the limb.

THE TEAM — E-JUST

Ahmed Elsheikh Karim Hossam Miral Farghaly Noran Morad Moustafa Abdullah
Ezaldeen Ahmed Rasha Gamel Mariam Ihab Noureen Tamer Jomana George

SUPERVISORS

Dr. Reda Albassiouny
Dr. Sameh Sherif